CSE1141 Fall2019

Programming Assignment #5

Due Date: 23/12/2019 – 23:00 (No Extension)

In this assignment, you will write the following programs by multidimensional arrays.

1. In this question, you will write a two-dimensional Grid Game. In this game, there is a person and multiple items. Items placed into the cell randomly. Person start cell point is (0,0) and cell status is 2. The aim of this game is the person collects all the items with limited number of movement. The user will move the person in the game with the right (R), left (L), down (D) and up (U) keys.

   Step 1: Your program should prompt the user to enter the size of Grid (***N***).
   Step 2: You will generate a two-dimensional ***(N)x(N) size*** grid area.
   Step 3: Generate ***N/2*** items and locate with randomly into the grid.
   Step 4: Define *maximum number of movement* as (***2.5)*N.***
   Step 5: Show to the user the current grid area and prompt the direction of the person (right (R), left (L), down (D) and up (U)). Use '***X'*** key as exit the game.
   Step 6: Move the person to the empty/item cell based on the direction and update the cell status (Previous cell status is 1, current cell status is 2). Person can't move to the already used cell.
   Step 7: If all items collected or there is no movement go to *Step 8*, otherwise go to *Step 6*.
   Step 8: Finish the game.

   Grid cell status:
   0: Empty cell not used
   1: Already used cell by person
   2: Current person position
   5: Items

   Console View:
   X: For person
   +: For used cells
   *: For items
   |: For left and right walls

   You have to check all inputs those are valid or invalid. If the inputs are invalid, you have to write appropriate messages.

   Here are some execution examples:
   Execution 1:

   1)
```
WELLCOME TO GRID GAME

Please enter the Grid size: 10
```

2)

```
--Hello 10x10 Grid Game--
|X          |
|     *     |
|           |
|     *   * |
|           |
|           |
|          *|
|   *       |
|           |
|           |
Moves:25        Current Cell:(0,0)
Choose Your Direction (R,L,U,D):R
```

3)

```
|+X         |
|     *     |
|           |
|     *   * |
|           |
|           |
|          *|
|   *       |
|           |
|           |
Moves:24        Current Cell:(0,1)
Choose Your Direction (R,L,U,D):D
```

4)

```
|++         |
| ++X*      |
|           |
|     *   * |
|           |
|           |
|          *|
|   *       |
|           |
|           |
Moves:21        Current Cell:(1,3)
Choose Your Direction (R,L,U,D):R
```

5)

```
|++         |
| ++++      |
|    +      |
|    +++++  |
|         + |
|         ++|
|          +|
|   *  X+++|
|           |
|           |
Moves:6 Current Cell:(7,6)
Choose Your Direction (R,L,U,D):L
```

6)

```
|++         |
| ++++      |
|    +      |
|    +++++  |
|         + |
|         ++|
|          +|
|   *X+++++|
|           |
|           |
Moves:4 Current Cell:(7,4)
Choose Your Direction (R,L,U,D):L


*** YOU WIN ***
```

Execution 2:

1)

```
WELLCOME TO GRID GAME

Please enter the Grid size: 12
```

2)

```
--Hello 12x12 Grid Game--
|X           |
| *          |
|            |
|            |
|            |
|      *     |
|            |
|            |
|           * |
|   **       |
|            |
|            |
|         *  |
Moves:30        Current Cell:(0,0)
Choose Your Direction (R,L,U,D):r
```

3)

```
|++          |
| +          |
| +          |
| +          |
| ++++++     |
|      +++++ |
|          + |
|          + |
|   ++++++++ |
|   +        |
|   X        |
|        *   |
Moves:3 Current Cell:(10,3)
Choose Your Direction (R,L,U,D):D
```

4)

```
|++          |
| +          |
| +          |
| +          |
| ++++++     |
|      +++++ |
|          + |
|          + |
|   ++++++++ |
|   +        |
|   +        |
|   +X   *   |
Moves:1 Current Cell:(11,4)
Choose Your Direction (R,L,U,D):R



-- GAME OVER --
```

2. In this question, you will write a program that tests whether a two-dimensional list has string keyword, either horizontally, vertically, or diagonally (forward→ or backward← and up↑ or down↓).

Step 1: Your program should prompt the user to enter the ***number of rows*** and ***columns*** of the list

Step 2: You will generate a two-dimensional list with random letters between 'A' to 'Z' and '0' to '9'.

Step 3: Print your two-dimensional list.

Step 4: Prompt the user to enter the ***keyword***.

Step 5: Your program then displays **True** if the list contains ***keyword***, either horizontally, vertically, or diagonally (forward→ or backward← and up↑ or down↓). Otherwise, it will display **False**.

Here are some examples of the true cases:

Keyword: *COMPUTER*	Keyword: *MARMARA*	Keyword: *2020*
G Z A 4 C 8 O 4 Z B P	Z K A 4 C 8 O 4 G B P	G Z A 4 C 8 O 4 Z B P
R Z K H A K F B N D F	N Z K H A K F B R D F	H Z K H A K F B N D F
E S X F 3 6 5 R T Q J	T M X F 3 6 5 R E Q J	E S X F 3 6 5 R T Q J
T 5 J Y G 3 2 N O Z T	O 5 A Y G 3 2 N T Z T	T 5 J Y G 3 2 N O Z T
U 5 E R N 6 E 1 K 2 R	K 5 E R N 6 E 1 U 2 R	0 5 E R N 6 E 1 K 2 R
P 6 C 2 X B J 5 T D U	T 6 C 2 M B J 5 P D U	P 6 C 2 0 2 0 5 T D U
M 4 I I M T I A Q F 3	Q 4 I I M A I A Z F 3	M 4 I I M T I A Q F 3
O 6 Q G B L 9 D V 8 K	V 6 Q G B L R D O 8 K	O 6 Q G B L 9 D V 8 K
C Q 9 A D M I K Q T E	Q Q 9 A D M I A C T E	9 Q 9 A D M I K Q T E
S R A W 7 V X 5 L P Q	L R A W 7 V X 5 S P Q	S R A W 7 V X 5 L P Q

It should be noted that you do not need to highlight the true case with yellow. We just add them to read it easily. Please do not forget to leave a space (' ') after printing each item in the list.

Here are some execution examples:

Execution 1:

```
Please enter the number of rows: 10
Please enter the number of columns: 10

 2 F S V 4 D N A N L
 8 O G I P 6 3 Q F E
 Y S E F C 6 K 2 C Z
 H K 2 B B V 0 N 2 W
 E P C 4 K 5 8 4 U T
 J H W L 0 I X A 5 B
 D V 5 B G K W 7 N Q
 Q 1 S S P C O S P O
 X U N Y R O L A H 5
 G C F T B B C J 8 X

Please enter the Keyword: BBC

TRUE
```

Execution 2:

```
Please enter the number of rows: 8
Please enter the number of columns: 12

 B 1 V J 5 1 T 6 M E S G
 H 7 I 5 Z U M C J L Q 8
 D W L N 1 8 S N 1 I 4 Z
 F O R P U M 4 J I U 9 G
 X 8 V R C 0 S X 0 W O D
 3 4 U H 5 F Q E J C T 4
 2 7 8 Y S A 2 E Y R M P
 T R 3 P G 2 F A B H H H

Please enter the Keyword: 2020

FALSE
```

Note: Your program should execute correctly for different test cases.

Submission Instructions:

Please zip and submit all your files using filename ***YourNumberHW5.zip*** (ex: 150713852HW5.zip) to Canvas system (under Assignments tab). Your zip file should contain the followings:

1. Java source code for Q1 (Pro5_1_150713852.java)
2. Java class file for Q1 (Pro5_1_150713852.class)
3. Java source code for Q2 (Pro5_2_150713852.java)
4. Java class file for Q2 (Pro5_2_150713852.class)

Your program ***must include necessary comments with your own words to explain your actions!***

Notes:

1. Write a comment at the beginning of each program to explain the purpose of the program.
2. Write your name and student ID as a comment.
3. Include necessary comments to explain your actions.
4. Select meaningful names for your variables.
5. You are allowed to use the materials that you have learned in lectures & labs.
6. Do not use the things that you did not learn in the course.
7. **Program submissions** should be done through the Canvas class page, under the assignments tab. Do not send program submissions through e-mail. E-mail attachments will not be accepted as valid submissions.
8. You are responsible for making sure you are turning in the right file, and that it is not corrupted in anyway. We will not allow resubmissions if you turn in the wrong file, even if you can prove that you have not modified the file after the deadline.
9. In case of any form of **copying and cheating** on solutions, all parts will get **ZERO** grade. You should submit your own work. In case of any forms of cheating or copying, both giver and receiver are equally culpable and suffer equal penalties.
   All types of plagiarism will result in zero grade from the homework.
10. No late submission will be accepted

Grading:

Problem 1 (45 points)

- Taking inputs (4 points)
- Creating a table of row-by-column values (10 points)
- Print the current position and move values (2 points)
- Positioning the Items (4 points)
- Movement (25)

Problem 2 (45 points)

- Taking inputs (6 points)
- Creating a table of row-by-column values (15 points)
- Correct execution for test inputs (24 points)
    - Finding keyword either
        - horizontally, forward or backward (8 points),
        - vertically, up or down (8 points),
        - diagonally forward or backward (8 points)

Submission format (10 points)

- 2 java files + 2 class files (4 points)
- Comments are necessary (6 points)
- Make sure that your class files can be executed on another computer.
- Make sure that the input/output of your program must be the same with the examples above (all informative strings & spaces).